Convergence

f L-smooth

$\downarrow$

Descent Lemma

$x^{k+1} = \arg\min_x \sim$

代入 x 为 x^k

$\downarrow$

Sufficient Descent Property

$$F^{k+1} + \frac{1}{2}\|\Delta x^k\|^2 \le F^k \xrightarrow{F \text{ bound}} x^k \text{ Bounded}$$

F coercive

$\downarrow$

$\downarrow \sum_{k=1}^{K} \|\Delta x^k\| \le F^k - F^0 < +\infty$

Wasserstrain

Interval $\Delta x^k \to 0$

Subseq $x^{k_i} \to \tilde{x}$

$\downarrow$

$x^{k+1} = \arg\min_x \sim$

代入 x 为 $\tilde{x}$ 代入 x^{k+1} 为 x^{k_i}

$$\xrightarrow{\limsup \ge} F(x^{k_i}) \to F(\tilde{x}) \xleftarrow{\liminf \ge} g \text{ LSC}$$

$\downarrow$

$x^{k+1} = \arg\min_x \sim$

Opt Cond. $0 \in \partial \sim$

$\tilde{x} \in \operatorname{Crit} F \leftarrow$ Gradient

$\|\partial F^k\| \to 0$

KL Property $\downarrow$

Seq $x^k \to \tilde{x}$